# Benchmarking Data-Integrity Anomaly Detection in Blood-Bank Transaction Records

Aryan Jain
*High School Graduate*
India

***Abstract*—Blood Establishment Computer Systems (BECS) operate under strict regulatory frameworks to ensure the traceability of the blood supply. While unsupervised machine learning (ML) is increasingly used for healthcare anomaly detection, the suitability of purely data-driven anomaly detection for discrete, domain-specific blood-bank integrity constraints remains less explored than its use for general statistical outlier detection. This paper evaluates a hybrid framework combining a deterministic rule engine with a Median Absolute Deviation (MAD) statistical detector against out-of-sample Isolation Forest (IF) and Local Outlier Factor (LOF) ML baselines. Using a synthetic taxonomy of 300 mutually exclusive anomalies injected across 2,000 records and evaluated over 10 independent train-test seed pairs, the hybrid framework achieved complete coverage of the predefined benchmark. In the latest experimental configuration, IF and LOF achieved mean F1 scores of 0.265 ± 0.045 and 0.367 ± 0.035, respectively. These results demonstrate the potential complementarity of deterministic validation and robust statistical detection for maintaining data integrity in highly regulated clinical environments.**

***Index Terms*—Data mining, anomaly detection, blood bank, health information systems, synthetic data.**

## I. Introduction

Modern blood banks rely on Blood Establishment Computer Systems (BECS) to systematically mitigate risks and maintain data integrity. BECS platforms must enforce validated workflows and time-stamped audit trails to maintain a reliable chain of custody [1].

Data mining techniques are frequently deployed to audit healthcare information systems, identifying transactional anomalies indicative of human error or software glitches [7]. Unsupervised machine learning (ML) models, such as Isolation Forests, are often favored due to their ability to detect novel patterns in high-dimensional data [4]. However, data-driven anomaly detectors may not directly encode discrete regulatory relationships, such as an issued unit lacking a valid donor reference.

This research frames its contributions around three primary areas:

1) A controlled synthetic benchmark containing distinct blood-bank data-integrity anomaly classes.
2) A robust out-of-sample comparison of deterministic and statistical detection, alongside IF and LOF baselines, evaluated by anomaly class over multiple independent seed generations.
3) A hybrid design implication demonstrating how rule-based and statistical methods can complement each other within a blood-bank validation architecture.

## II. Regulatory Framework and Traceability

The architectural requirements for data integrity in blood banking are dictated by regulatory mandates. WHO guidance emphasizes the importance of traceability, documentation, and controlled records in blood establishments [1]. Traceability requires systems capable of forward and backward tracking, linking the donor's intake profile directly to final disposition [8].

In the Indian clinical context, biological tracking requirements are codified by the Drugs and Cosmetics Act of 1940 and its subsequent Rules [2]. Blood centers must maintain exhaustive records for processing human blood, ensuring the chain of custody remains unbroken. Global systems often implement standards like ISBT 128 to provide international consistency for barcode labeling and traceability [3].

From a data-mining perspective, ensuring compliance requires verifying that clinical data conforms to a complex state machine. For the purposes of this benchmark, the blood-unit lifecycle is represented conceptually as

$$\text{COLLECTED} \rightarrow \text{TESTED} \rightarrow \text{AVAILABLE} \rightarrow \text{ISSUED}.$$

Invalid transitions, such as COLLECTED $\rightarrow$ ISSUED (bypassing mandatory testing), are anomalous structural constraints that systems must enforce. Similarly, an issued unit with a reactive test result represents an invalid lifecycle condition in the synthetic benchmark.

## III. Algorithmic Framework and Data Engineering

Due to the sensitive nature of protected health information, a synthetic dataset with known ground truth was generated to facilitate reproducible benchmarking [6].

### A. Data Synthesis and Population Modeling

The engineering process initialized a registry of 500 synthetic donors. Blood-group assignments were generated using fixed probabilities chosen to produce a heterogeneous synthetic donor population. Using this registry, 2,000 unit transactions were simulated over a 90-day operational window. The units were partitioned into four clinical components with specific regulatory shelf lives: Packed Red Blood Cells (42 days), Whole Blood (35 days), Fresh Frozen Plasma (365 days), and Platelets (5 days). Clean data generation mathematically guarantees zero internal constraints are violated prior to injection.

### B. Controlled Error Taxonomy

To establish a rigorous mathematical benchmark, exactly 300 anomalies (15% anomaly ratio) were injected into the synthetic dataset. The injected anomalies were mutually exclusive at the transaction level, such that each anomalous transaction was assigned to exactly one taxonomic class. Duplicate injection mechanisms were constrained to copy from verified clean records to preserve this mutual exclusivity. This taxonomy encompasses nine distinct failure modes:

- **Missing Fields (n=34):** Mandatory data points removed, including categorical and timestamp fields.
- **Duplicate Records (n=34):** Unit identifier collisions.
- **Conflicting Records (n=34):** Unit blood group contradicts the donor's historical data, simulating a transaction-level entry error.
- **Impossible States (n=33):** Units issued despite reactive/discard tests.
- **Expiry Violations (n=33):** Issue timestamps succeeding the biological expiration limit.
- **Traceability Breaks (n=33):** Transaction references a donor identifier that cannot be resolved against the registry.
- **Format Inconsistencies (n=33):** Free-text corruptions of constrained enumerations.
- **Temporal Anomalies (n=33):** Issue timestamps chronologically prior to collection timestamps.
- **Extreme Univariate Numerical Deviations (n=33):** Extreme synthetic weight (188.5 kg) and hemoglobin (24.8 g/dL) outliers.

### C. Algorithmic Implementations

The benchmark evaluates three individual detection approaches and a hybrid integration.

**Rule-Based Validator:** A deterministic domain-integrity validator programmed with strict operational logic. It parses each transaction against completeness, domain integrity, identifier uniqueness, referential integrity, issued-state constraints, and temporal/expiry logic. The validator has access to the authoritative synthetic donor registry, representing a domain-informed validation setting.

**Statistical Anomaly Detector:** To capture numerical outliers, a statistical module utilized the Modified Z-Score based on the Median Absolute Deviation (MAD) [5]. MAD parameters were calibrated exclusively on the clean training set before anomaly injection. Observations yielding a modified Z-score $> 3.5$ for the modeled continuous numerical variables on the test set were flagged as anomalous.

**Machine Learning Baselines (IF and LOF):** An Isolation Forest [4] and a Local Outlier Factor (LOF) model [9] were deployed as unsupervised feature-space anomaly-detection baselines with contamination $c = 0.15$. Input features comprised donor weight, donor hemoglobin, quantity issued, collection-to-issue lead time, and one-hot encoded blood group, state, test status, component, and storage-location variables. To prevent feature-space mismatch between independent splits, a global OneHotEncoder was fitted to a predefined vocabulary. Missing fields and malformed categorical values were represented explicitly as distinct categorical states.

**Hybrid Detector:** The final anomaly decision was defined as the logical OR of the rule-based and MAD detector outputs. A transaction was classified as anomalous if either deterministic validation or MAD-based outlier detection flagged it.

To ensure generalizability, 10 independent train and test seed pairs were generated. ML models and MAD parameters were fitted on training data and evaluated out-of-sample (OOS) on independent test datasets. The experimental script automatically exports overall, multi-seed, per-class, and sensitivity metrics as CSV artifacts.

TABLE I
OVERALL OUT-OF-SAMPLE BENCHMARK RESULTS (MEAN $\pm$ SD)

| Method | Precision | Recall | F1 | FPR |
|---|---|---|---|---|
| Rule | 1.000 $\pm$ 0.000 | 0.890 $\pm$ 0.000 | 0.942 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| MAD | 1.000 $\pm$ 0.000 | 0.110 $\pm$ 0.000 | 0.198 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |
| IF | 0.262 $\pm$ 0.046 | 0.269 $\pm$ 0.046 | 0.265 $\pm$ 0.045 | 0.134 $\pm$ 0.014 |
| LOF | 0.348 $\pm$ 0.037 | 0.390 $\pm$ 0.041 | 0.367 $\pm$ 0.035 | 0.130 $\pm$ 0.018 |
| Hybrid | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 1.000 $\pm$ 0.000 | 0.000 $\pm$ 0.000 |

## IV. EMPIRICAL BENCHMARKING AND RESULTS

Table I presents the overall out-of-sample classification performance averaged across the 10 independent seed pairs. The latest experimental configuration includes component and storage-location categorical features and the corrected mandatory-field check for `expiry_timestamp`.

The Rule-Based Validator achieved perfect precision on the deterministic constraints but missed the numerical burst class. The MAD detector identified the extreme univariate numerical deviations but bypassed the structural violations. The Hybrid model captured all 300 predefined anomalies.

The MAD magnitude sensitivity experiment further showed that the detector's effectiveness depends on the magnitude of the injected numerical deviation: the extreme class was detected completely, whereas moderate and subtle deviations were not reliably captured. The experimental code also contains an out-of-sample Isolation Forest contamination sensitivity analysis; its results are retained in the generated CSV artifact rather than reproduced here, avoiding the mixing of results from the previous feature configuration with the current manuscript version.

Table II shows that the deterministic validator achieved complete recall across the seven structural classes represented by its rule set, while MAD captured the extreme numerical class. IF and LOF were inconsistent across classes. LOF substantially outperformed IF on expiry and temporal anomalies, while both methods performed strongly on the deliberately extreme numerical class.

TABLE II
PER-CLASS RECALL PERCENTAGES (MEAN $\pm$ SD, %)

| **Anomaly Class** | **Rule** | **MAD** | **IF** | **LOF** |
|---|---|---|---|---|
| Missing Fields | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $19.4 \pm 6.7$ | $14.7 \pm 4.4$ |
| Duplicate Records | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $12.4 \pm 3.7$ | $11.5 \pm 3.3$ |
| Conflicting Records | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $16.8 \pm 7.1$ | $10.9 \pm 3.7$ |
| Impossible States | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $77.0 \pm 20.6$ | $78.5 \pm 10.9$ |
| Expiry Violations | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $19.7 \pm 6.9$ | $42.7 \pm 23.8$ |
| Traceability Breaks | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $8.8 \pm 5.0$ | $15.2 \pm 5.4$ |
| Format Inconsistencies | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $14.5 \pm 5.2$ | $11.5 \pm 6.2$ |
| Temporal Anomalies | $100.0 \pm 0.0$ | $0.0 \pm 0.0$ | $3.0 \pm 2.7$ | $69.7 \pm 14.1$ |
| Extreme Univariate | $0.0 \pm 0.0$ | $100.0 \pm 0.0$ | $71.2 \pm 20.1$ | $98.5 \pm 3.6$ |

## V. DISCUSSION AND LIMITATIONS

The empirical results reveal that different anomaly classes exhibit distinct mathematical structures. Deterministic constraints are naturally captured by explicit rules, while extreme univariate numerical deviations can be captured by robust statistical methods. The tested ML configurations struggled because structural violations, such as unresolved donor identifiers or missing categorical fields, are not necessarily represented as large deviations in the feature space supplied to IF and LOF.

A false-positive analysis on the primary $42 \rightarrow 100$ split showed that IF generated 274 false positives among 1,700 clean records. In that clean population, O+ represented 32.2% of records and 39.8% of the false-positive flags; B+ represented 28.9% and 22.3%, respectively; A+ represented 26.2% and 21.2%; and AB+ represented 6.9% and 9.1%. This analysis is illustrative of the distinction between statistical rarity and domain invalidity rather than evidence that any particular blood group is inherently anomalous.

Several limitations should be emphasized. First, all data were synthetically generated and the anomaly classes were deliberately engineered. Second, the numerical burst class used deliberately extreme values, so the strong MAD result should not be interpreted as evidence of sensitivity to subtle clinical drift. Third, the contamination parameter supplied to IF and LOF is a model assumption that influences the precision-recall trade-off. Fourth, the deterministic validator has access to the authoritative synthetic donor registry, representing a domain-informed validation setting rather than a purely data-driven detector. Finally, the 100% hybrid score reflects complete coverage of the predefined synthetic taxonomy and should not be interpreted as validation of a clinical deployment system.

## VI. CONCLUSIONS

This benchmark demonstrates complementary coverage of predefined structural and extreme numerical anomaly classes by deterministic and robust statistical detectors, evaluated out-of-sample across 10 independent synthetic train-test generations. The tested IF and LOF baselines showed substantially lower and more variable performance than the deterministic validator, particularly on discrete integrity constraints, although their class-level behavior was not identical. MAD was effective for the deliberately extreme univariate anomaly class but insensitive to the moderate and subtle deviations tested. The findings support further evaluation of hybrid validation architectures using independently collected or appropriately de-identified real-world blood-bank transaction data and richer representations of relational and temporal structure.

## REFERENCES


[1] World Health Organization, “WHO guidelines on good manufacturing practices for blood establishments,” *WHO Technical Report Series*, Annex 4, TRS No. 961, 2011.

[2] Government of India, “Drugs and Cosmetics Act, 1940 and Rules, 1945,” Ministry of Health and Family Welfare, India, 1945.

[3] ICCBBA, “ISBT 128 Standard Technical Specification,” ICCBBA, 2024.

[4] F. T. Liu, K. M. Ting, and Z.-H. Zhou, “Isolation-based anomaly detection,” *ACM Transactions on Knowledge Discovery from Data*, vol. 6, no. 1, pp. 1–39, 2012.

[5] C. Leys et al., “Detecting outliers: Do not use standard deviation around the mean, use absolute deviation around the median,” *Journal of Experimental Social Psychology*, vol. 49, no. 4, pp. 764–766, 2013.

[6] R. J. Chen et al., “Synthetic data in machine learning for medicine and healthcare,” *Nature Biomedical Engineering*, vol. 5, pp. 493–497, 2021.

[7] M. A. F. Pimentel et al., “A review of novelty detection,” *Signal Processing*, vol. 99, pp. 215–249, 2014.

[8] C. H. Jenkins et al., “Blood transfusion traceability and vein-to-vein tracking,” *Transfusion Medicine Reviews*, vol. 35, no. 1, pp. 1–8, 2021.

[9] M. M. Breunig, H.-P. Kriegel, R. T. Ng, and J. Sander, “LOF: Identifying density-based local outliers,” *ACM SIGMOD Record*, vol. 29, no. 2, pp. 93–104, 2000.